

Google Cloud Platform (GCP) is a suite of cloud computing services that runs on the same infrastructure that Google uses internally. Highly Scalable, Secure, Reliable & Global.

KEY FEATURES

- Global Infrastructure
- High Scalability & Performance
- Security & Identity Management
- Pay-as-you-go Pricing
- Big Data, AI/ML, Analytics
- DevOps & Developer Friendly

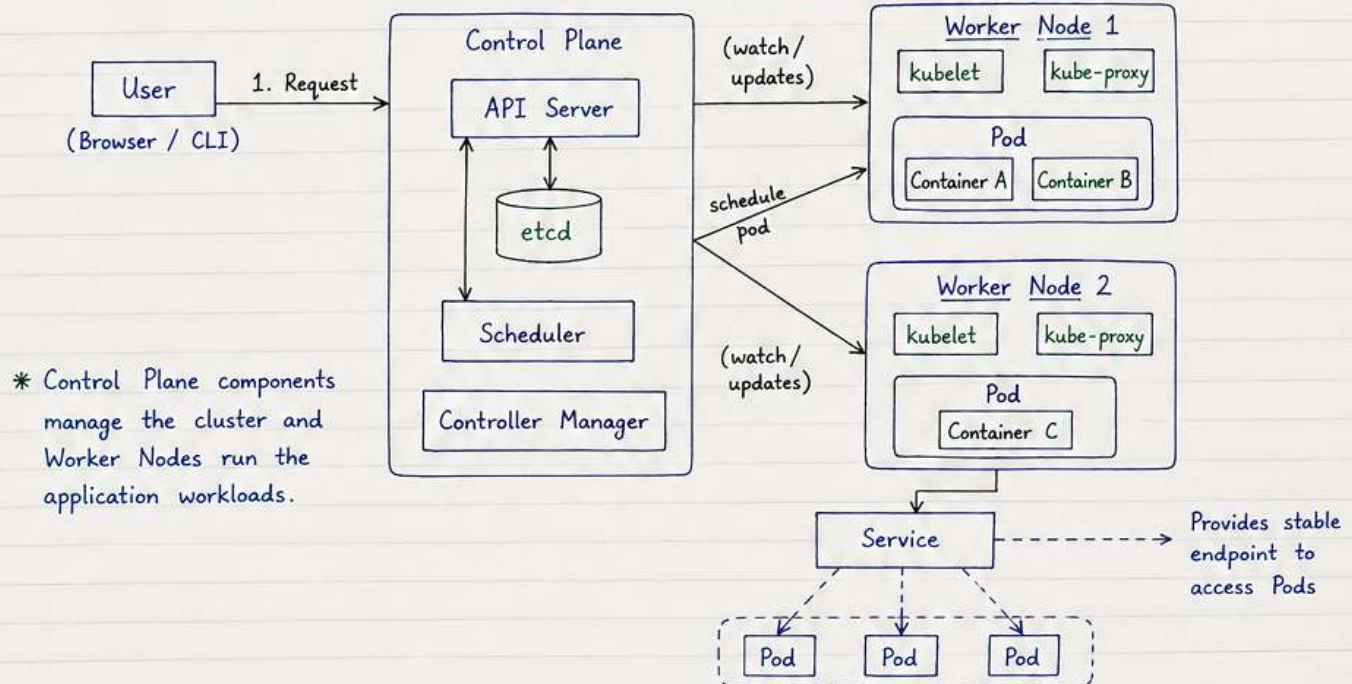
GCP GLOBAL INFRASTRUCTURE

- Regions → Geographic areas
- Zones → Isolated locations in a region
- Multi-Region → High availability
- Edge Locations → For CDN & caching

PRICING MODEL

- Pay-as-you-go
- Sustained Use Discounts
- Committed Use Discounts
- Free Tier (limited services)

GCP ARCHITECTURE (HIGH LEVEL)



HOW IT WORKS (Flow)

1. User runs kubectl command.
2. API Server receives request.
3. Scheduler selects Worker Node.
4. kubelet creates Pod.
5. Container starts & app is running.
6. etcd stores cluster state.

API Server : Frontend for the Kubernetes control plane.

etcd : Consistent and highly-available key-value store for all cluster data.

Scheduler : Assigns Pods to suitable Worker Node.

Controller Manager : Runs controllers (Node, ReplicaSet, Endpoints, Namespace, etc.)

kubelet : Agent that runs on each node and ensures containers are running.

kube-proxy : Handles networking & service load balancing.

IMPORTANT GCP SERVICES (CATEGORIES)

COMPUTE	STORAGE	NETWORKING	DATABASE	ANALYTICS	AI / ML
• Compute Engine • GKE (Kubernetes) • Cloud Functions • App Engine	• Cloud Storage • Persistent Disk • Cloud SQL • Filestore	• VPC • Cloud Load Balancing • Cloud CDN • Cloud DNS	• Cloud SQL • Cloud Spanner • Firestore • Bigtable	• BigQuery • Dataflow • Dataproc • Pub/Sub	• Vertex AI • AutoML • AI Platform • Vision / NLP APIs

SECURITY & IAM

- IAM (Identity & Access Management)
- Service Accounts
- Cloud Security Command Center
- VPC Service Controls

DEVOPS & MANAGEMENT

- Cloud Monitoring
- Cloud Logging
- Cloud Build
- Deployment Manager

Remember :
Understand Concepts
Practice Diagrams
You've Got This! 😊

EXAM / INTERVIEW TIP

- Focus on Core Services
- Understand Architecture
- Use Diagrams
- Hands-on Practice



COMPUTE SERVICES

- Compute Engine - VMs in the cloud
- GKE (Kubernetes) - Managed Kubernetes
- App Engine - PaaS to build & deploy apps
- Cloud Run - Run containers, fully managed
- Cloud Functions - Serverless functions
- Cloud Batch - Batch processing jobs

STORAGE SERVICES

- Cloud Storage - Object storage
- Persistent Disk - Block storage for VM
- Filestore - Managed NFS file storage
- Cloud SQL - Managed relational DB
- Cloud Spanner - Globally distributed DB
- Bigtable - NoSQL wide column DB

NETWORKING SERVICES

- VPC - Isolated private cloud
- Cloud Load Balancing - Distribute traffic
- Cloud CDN - Global content delivery
- Cloud DNS - Managed DNS service
- Cloud Armor - DDoS protection
- Cloud Interconnect - Hybrid connectivity

IAM (IDENTITY & ACCESS MANAGEMENT)

- Manage access to GCP resources securely
- Principle of Least Privilege
- Components:
 - Identity - Users, Service Accounts, Groups
 - Role - Set of permissions
 - Policy - Bindings of role to identity
- Use: gcloud iam / Console / API

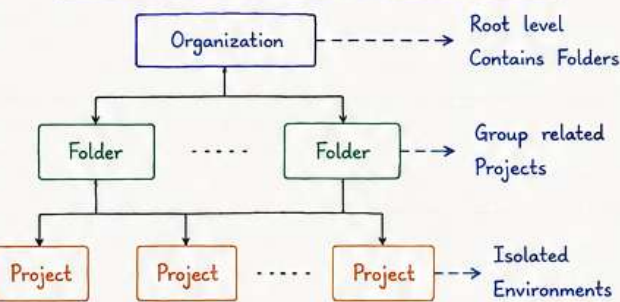
SECURITY SERVICES

- Cloud Security Command Center - Security posture
- Identity-Aware Proxy (IAP) - Secure access to apps
- Cloud Armor - DDoS & WAF protection
- VPC Service Controls - Data exfiltration protection
- Secret Manager - Store & manage secrets
- Encryption - Data in transit & at rest

DEVOPS & MANAGEMENT

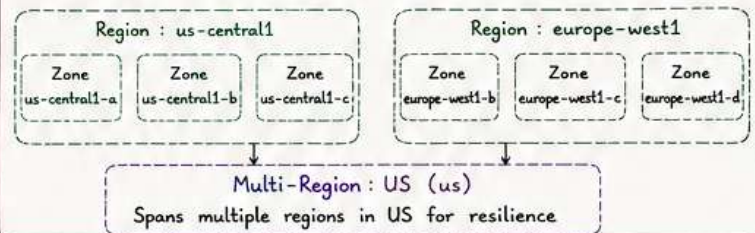
- Cloud Monitoring - Metrics, logs, alerts
- Cloud Logging - Centralized logging
- Cloud Build - Build, test & deploy
- Artifact Registry - Store container images
- Deployment Manager - IaC with templates
- Pub/Sub - Messaging service

GCP PROJECTS, FOLDERS & ORGANIZATIONS



REGIONS, ZONES & MULTI-REGION

- Region - A specific geographic location (e.g. us-central1)
- Zone - Isolated location within a region (e.g. us-central1-a)
- Multi-Region - Spans multiple regions for high availability



GCP PRICING SUMMARY

Model	Description
Pay-as-you-go	Pay for what you use, per second/minute/hour.
Sustained Use Discounts	Automatic discounts for running resources for a significant portion of the month.
Committed Use Discounts	Commit to specific usage for 1 or 3 years in exchange for lower rates.
Free Tier	Always free resources up to limits.
Preemptible / Spot VMs	Use spare capacity at a lower price.

HIGH AVAILABILITY & SCALABILITY

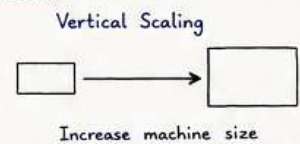
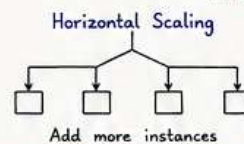
High Availability

- Use multiple zones in a region
- Load Balancers distribute traffic
- Health Checks detect failures
- Auto-healing for instances

Scalability

- Horizontal Scaling - Add more instances
- Vertical Scaling - Increase machine size
- Autoscaler - Scale automatically based on demand

TYPES OF SCALING



GCP DATA & ANALYTICS

- BigQuery - Serverless data warehouse
- Dataflow - Stream & batch data processing
- Dataproc - Managed Hadoop, Spark, Kafka
- Pub/Sub - Real-time messaging
- Data Studio (Looker Studio) - Data visualization

AI / ML SERVICES

- Vertex AI - Unified ML platform
- AutoML - Build ML models with little code
- Vision AI - Image analysis
- Natural Language API - NLP tasks
- Speech-to-Text / Text-to-Speech

MIGRATION & HYBRID CLOUD

- Cloud VPN / Interconnect - Secure connectivity
- Anthos - Run apps across on-prem & multi-cloud
- Migrate for Compute Engine - Lift & shift
- Database Migration Service - Migrate databases
- Transfer Appliance - Large data transfer

IMPORTANT GCP COMMANDS (gcloud)

- gcloud auth login - Login to GCP
- gcloud config set project PROJECT_ID - Set project
- gcloud compute instances list - List VM instances
- gcloud sql instances list - List Cloud SQL instances
- gcloud container clusters list - List GKE clusters
- gcloud storage ls - List Cloud Storage buckets
- gcloud iam roles list - List IAM roles

LAST MINUTE TIPS

- ✓ Focus on Core Services
- ✓ Practice Diagrams
- ✓ Understand Use Cases
- ✓ Learn IAM & Security
- ✓ Hands-on in Console / Qwiklabs

INTERVIEW / EXAM TIPS

- ☑ Understand Architecture & Components
- ☑ Know use-cases of each service
- ☑ Draw neat diagrams
- ☑ Compare services (e.g. App Engine vs Cloud Run)
- ☑ Practice Hands-on





COMPUTE SERVICES (Details)



- Compute Engine - IaaS VMs, customizable, scalable virtual machines.
- GKE (Google Kubernetes Engine) - Managed Kubernetes service for containers.
- App Engine - PaaS to build & run web apps without managing infrastructure.
- Cloud Run - Serverless platform to run containers in response to requests.
- Cloud Functions - Event-driven serverless compute (small pieces of code).
- Cloud Batch - Run batch jobs at scale.

STORAGE SERVICES (Details)



- Cloud Storage - Store unstructured data (images, videos, backups, logs).
- Persistent Disk - Block storage for Compute Engine VMs.
- Filestore - Fully managed NFS file storage.
- Cloud SQL - Managed MySQL, PostgreSQL, SQL Server databases.
- Cloud Spanner - Globally distributed, strongly consistent relational DB.
- Bigtable - NoSQL wide column database for large scale workloads.

NETWORKING SERVICES (Details)



- VPC - Virtual Private Cloud for isolated network.
- Cloud Load Balancing - Distribute traffic across instances.
- Cloud CDN - Cache content closer to users.
- Cloud DNS - Managed DNS service.
- Cloud Armor - Protect against DDoS & web attacks.
- Cloud Interconnect - Dedicated connectivity between on-prem & GCP.

IAM - ROLES (Common Roles)

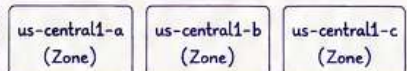


Role	Description
Owner	Full access to all resources.
Editor	Modify resources but not IAM & billing.
Viewer	Read-only access to resources.
Compute Admin	Manage Compute Engine resources.
Storage Admin	Manage Cloud Storage resources.
Security Admin	Manage security settings and policies.
Billing Admin	Manage billing account and budgets.

REGION vs ZONE

- Region - Large geographic area containing one or more zones.
- Zone - Isolated location within a region with separate power, networking.

Example : us-central1 (Region)



Deploy resources in multiple zones for High Availability.

RESOURCE HIERARCHY



GCP SERVICES BY USE CASE ☆

Use Case	Recommended Services
Web / App Hosting	App Engine, Cloud Run, Compute Engine, GKE
Databases	Cloud SQL, Cloud Spanner, Firestore, Bigtable
Data Storage	Cloud Storage, Persistent Disk, Filestore
Analytics / Big Data	BigQuery, Dataflow, Dataproc, Pub/Sub
Machine Learning	Vertex AI, AutoML, AI Platform
DevOps / CI/CD	Cloud Build, Artifact Registry, Deployment Manager
Monitoring / Logging	Cloud Monitoring, Cloud Logging, Error Reporting
Security	IAM, VPC Service Controls, Cloud Armor, KMS

COMMON USE CASES

- Host a Website → Cloud Run / App Engine
- Run Microservices → GKE / Cloud Run
- Store Files & Backups → Cloud Storage
- Run Databases → Cloud SQL / Spanner / Firestore
- Data Warehouse → BigQuery
- Real-time Streaming → Pub/Sub, Dataflow
- CI/CD Pipelines → Cloud Build
- Monitor & Alert → Cloud Monitoring
- Secure Applications → IAM, Cloud Armor



IMPORTANT GCP COMMANDS (Continued)



- gcloud projects list - List all projects
- gcloud config set project PROJECT_ID - Set default project
- gcloud compute instances list - List VM instances
- gcloud compute instances create VM_NAME --zone=ZONE - Create VM
- gcloud compute instances delete VM_NAME --zone=ZONE - Delete VM
- gcloud auth list - List authenticated accounts
- gcloud config list - Show configuration
- gcloud services enable SERVICE_NAME - Enable API
- gcloud services list --enabled - List enabled services
- gcloud iam roles list - List available IAM roles
- gcloud storage buckets list - List storage buckets
- gcloud beta container clusters list --zone=ZONE - List GKE clusters
- gcloud logging read "resource.type=gce_instance" - Read logs

GCP FREE TIER HIGHLIGHTS



- \$300 free credits for 90 days
- Always Free products (limited usage):
 - Compute Engine (1 f1-micro VM / month)
 - Cloud Storage (5 GB standard storage)
 - Cloud SQL (SQLite, MySQL small instance)
 - BigQuery (1 TB queries / month)
 - Cloud Functions (2M invocations / month)
 - Cloud Run (2M requests / month)

Note: Limits & regions apply.

Check latest pricing on GCP website.

QUICK TIPS



- ✓ Choose correct Region & Zone for better performance.
- ✓ Use Managed Services to reduce operational overhead.
- ✓ Enable Billing Alerts to avoid unexpected charges.
- ✓ Use IAM least privilege principle.
- ✓ Monitor usage with Cloud Monitoring & Budgets.

INTERVIEW / EXAM TIPS



- Understand core services & their use cases.
- Know the global infrastructure (Regions, Zones).
- Be clear on IAM, Security & Networking.
- Practice gcloud basic commands.
- Draw architecture diagrams.
- Compare services (Compute, Storage, Databases).
- Read official docs for updates.

LAST MINUTE REVISION CHECKLIST



- ✓ Core Services (Compute, Storage, Networking)
- ✓ IAM - Roles & Best Practices
- ✓ Regions, Zones & Multi-Region
- ✓ Pricing Models & Free Tier
- ✓ gcloud Commands
- ✓ Use Cases of each service
- ✓ Architecture Diagrams Practice

KEY TAKEAWAY



GCP provides Powerful, Scalable, Secure & Cost-effective cloud solutions.

Focus on core concepts, practice & real-world use cases - You're Ready!



GCP NETWORKING (Key Points)

- VPC is the foundation of GCP network.
- Subnet works in a specific region.
- Firewall rules control traffic to/from instances.
- Cloud Load Balancing distributes traffic across backends.
- Cloud CDN caches content at edge locations.
- Cloud NAT allows instances without external IP to access internet.

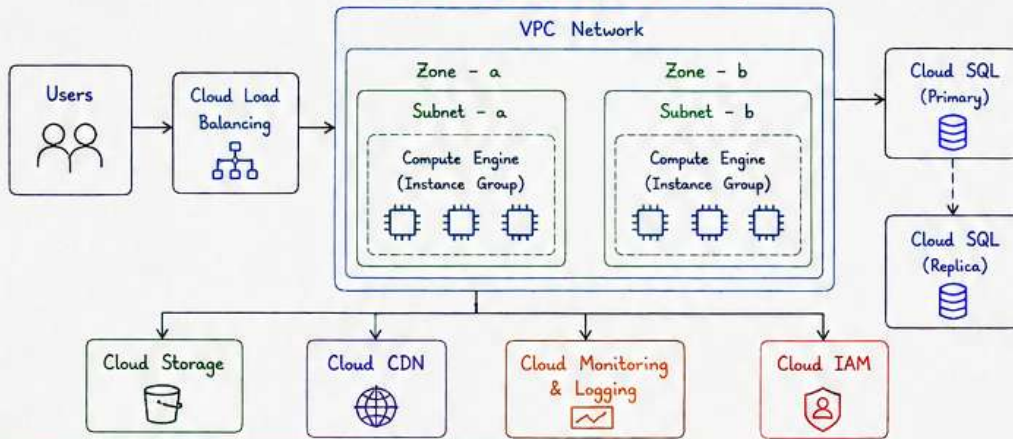
STORAGE OPTIONS (Quick View)

- Cloud Storage - Object storage for unstructured data.
- Persistent Disk - Block storage for VMs.
- Filestore - Managed NFS for lift & shift.
- Cloud SQL - Managed relational DB.
- Cloud Spanner - Globally distributed relational DB.
- Bigtable - NoSQL wide column store.
- Firestore / Datastore - NoSQL document DB.

SECURITY BEST PRACTICES

- Follow IAM least privilege principle.
- Use Service Accounts for applications.
- Enable MFA for users.
- Encrypt data at rest and in transit.
- Use VPC Service Controls for data exfiltration protection.
- Regularly review IAM roles and audit logs.
- Use Cloud Armor to protect applications from DDoS & attacks.

GCP ARCHITECTURE OVERVIEW



HIGH AVAILABILITY

- Use multiple zones in a region.
- Use Regional / Multi-Regional resources.
- Health checks + Auto-healing.
- Load Balancing across zones.
- Database replication.
- Backups and disaster recovery.

COMPUTE OPTIONS

- Compute Engine - Scalable VMs.
- GKE (Google Kubernetes Engine) - Managed Kubernetes.
- Cloud Run - Serverless containers.
- App Engine - PaaS for web apps.
- Cloud Functions - Serverless functions.
- Cloud Batch - Run batch jobs.

DATABASE OPTIONS

- Cloud SQL - MySQL, PostgreSQL, SQL Server.
- Cloud Spanner - Globally distributed SQL DB.
- Bigtable - NoSQL wide column family store.
- Firestore - NoSQL document database.
- Memorystore - Managed Redis / Memcached.
- AlloyDB - Managed PostgreSQL compatible.

DEVOPS & MANAGEMENT TOOLS

- Cloud Build - Build, test & deploy.
- Artifact Registry - Store container images.
- Deployment Manager / Terraform - IaC.
- Cloud Monitoring - Metrics & alerts.
- Cloud Logging - Centralized logs.
- Error Reporting - Track & fix errors.

GCP PRICING MODELS

- Pay-as-you-go - Pay for what you use.
- Sustained Use Discounts - Automatic discounts for frequently used resources.
- Committed Use Discounts - Commit for 1 or 3 years for bigger savings.
- Preemptible / Spot VMs - Use spare capacity at lower cost.
- Free Tier - Limited free resources.

GCP GLOBAL INFRASTRUCTURE

- Regions → Geographic areas (us-central1, europe-west1, asia-south1, etc.)
- Zones → Isolated locations within a region (us-central1-a, us-central1-b, us-central1-c)
- Multi-Region → Spans multiple regions for high availability (e.g., US, EU).
- Edge Locations → For CDN & caching.

IMPORTANT GCP CONCEPTS

- Project → Logical grouping of resources.
- Resource → Any service or component.
- IAM → Who can do What on Which resource.
- Region → Location for resources.
- Zone → Data center within a region.
- Quota → Limits for resources.
- Labels → Key-Value pairs to organize resources.

GCP KEY SERVICES (Cheat Sheet)

Compute Engine	Virtual machines in the cloud
GKE	Managed Kubernetes service
Cloud Run	Run containers without managing servers
App Engine	Platform to build & deploy apps
Cloud Storage	Object storage for any type of data
Cloud SQL	Managed MySQL, PostgreSQL, SQL Server
Cloud Spanner	Globally distributed relational database
Bigtable	NoSQL wide column database
Pub/Sub	Messaging service for event-driven systems
Cloud Functions	Event-driven serverless functions
Cloud CDN	Content delivery network
Cloud Load Balancing	Distribute traffic across instances
Cloud Armor	DDoS & WAF protection
Cloud Monitoring	Monitor metrics, uptime & alerts
Cloud Logging	Collect & analyze logs

TYPES OF DEPLOYMENT

- Lift & Shift
→ Move applications to cloud as-is.
- Modernize
→ Improve applications using managed services.
- Re-architect
→ Redesign for cloud-native & serverless.
- Rebuild
→ Rebuild applications for cloud benefits.
- Repurchase
→ Replace with SaaS products.

LAST MINUTE TIPS

- ✓ Understand core services & their use cases.
- ✓ Know Regions, Zones & Multi-Region.
- ✓ Learn IAM roles & best practices.
- ✓ Practice gcloud basic commands.
- ✓ Focus on networking & security.
- ✓ Understand pricing models.
- ✓ Draw architecture diagrams.
- ✓ Solve use cases & scenario-based questions.

INTERVIEW / EXAM TIPS

- ✓ Read the question carefully.
- ✓ Start with high level architecture.
- ✓ Mention services & data flow.
- ✓ Consider security, scalability, HA & cost.
- ✓ Conclude with monitoring & best practices.

NETWORKING CONCEPTS (More)



- **VPC Peering** - Private connectivity between VPC networks.
- **Cloud VPN** - Secure connection between on-prem & GCP over the internet.
- **Cloud Interconnect** - Dedicated private connection for high bandwidth & low latency.
- **Cloud NAT** - Allows instances in private subnet to access internet.
- **VPC Service Controls** - Create a security perimeter around services.

CLOUD IAM BEST PRACTICES



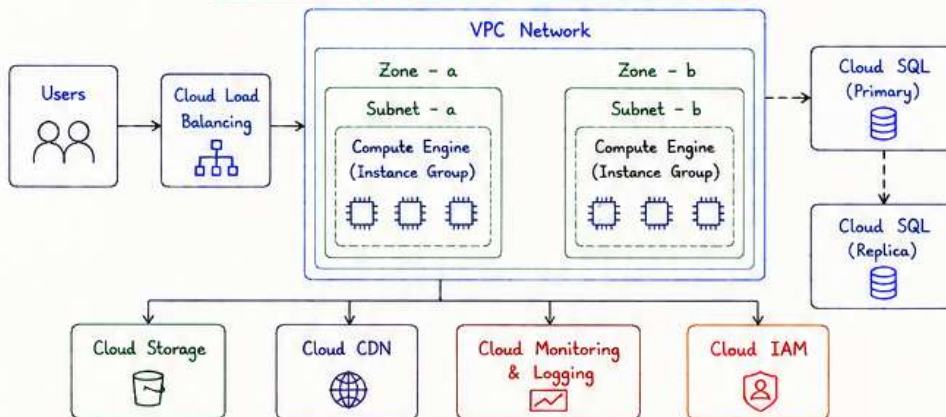
- Grant only the minimum access (Least Privilege).
- Use predefined roles whenever possible.
- Create custom roles for fine-grained access.
- Use Service Accounts for applications, not user accounts.
- Enable IAM Conditions for context-aware access.
- Regularly audit IAM policies & remove unused permissions.

CLOUD SQL & DATABASES



- **Cloud SQL** - Managed relational DB (MySQL, PostgreSQL, SQL Server).
- **Cloud Spanner** - Globally distributed relational database.
- **Firestore** - NoSQL document database.
- **Bigtable** - NoSQL wide-column database.
- **Memorystore** - Managed Redis / Memcached.
- **AlloyDB** - PostgreSQL compatible, high performance.

GCP ARCHITECTURE DIAGRAM (AT A GLANCE)



DISASTER RECOVERY STRATEGIES

- Use Multi-Region or Regional resources.
- Setup Backup schedules (Cloud SQL, Firestore, GCS).
- Replicate critical data across regions.
- Use Cloud Storage Nearline/Coldline for cheaper backups.
- Test DR plan regularly.
- Define RPO & RTO.



CLOUD STORAGE CLASSES



- **Standard** - Frequently accessed data.
- **Nearline** - Infrequent access (30 days min).
- **Coldline** - Rarely accessed data (90 days min).
- **Archive** - Long-term backup (365 days min).
- **Autoclass** - Automatically moves data to appropriate class.

MONITORING & OBSERVABILITY



- **Cloud Monitoring** - Metrics, dashboards, alerts.
- **Cloud Logging** - Centralized logs management.
- **Error Reporting** - Detect & fix errors.
- **Trace** - Request latency tracing.
- **Uptime Checks** - Monitor availability.

SECURITY SERVICES



- **Cloud Armor** - DDoS protection & WAF.
- **Security Command Center** - Risk visibility & security posture.
- **Binary Authorization** - Ensure trusted images.
- **Secret Manager** - Store & manage secrets.
- **KMS** - Encrypt & manage encryption keys.

COMPUTE SERVICES



- **Compute Engine** - IaaS virtual machines.
- **GKE** - Managed Kubernetes service.
- **Cloud Run** - Serverless containers.
- **App Engine** - PaaS for web apps.
- **Cloud Functions** - Event-driven serverless.
- **Cloud Batch** - Run batch jobs at scale.
- **Instance Groups** - Auto scaling, self-healing.
- **Sole-tenant Nodes** - Dedicated hardware.

SERVERLESS OPTIONS



- **Cloud Run** - Run containers without managing infra.
- **Cloud Functions** - Run code in response to events.
- **Firebase Functions** - Serverless with Firebase.
- **App Engine (Standard/Flexible)** - PaaS platform.
- **Eventarc** - Route events to services.

DATA & ANALYTICS



- **BigQuery** - Serverless data warehouse.
- **Dataflow** - Stream & batch processing.
- **Dataproc** - Managed Hadoop / Spark.
- **Pub/Sub** - Real-time messaging.
- **Looker / Looker Studio** - BI & dashboards.
- **Vertex AI** - ML platform for training & deployment.

IMPORTANT GCP COMMANDS (gcloud)



- | | |
|--|-------------------------|
| gcloud auth login | - Login to GCP |
| gcloud config set project PROJECT_ID | - Set project |
| gcloud projects list | - List all projects |
| gcloud compute instances list | - List VM instances |
| gcloud compute instances create NAME --zone=ZONE | - Create VM |
| gcloud compute instances delete NAME --zone=ZONE | - Delete VM |
| gcloud storage buckets list | - List all buckets |
| gcloud storage buckets create BUCKET_NAME | - Create bucket |
| gcloud iam roles list | - List all IAM roles |
| gcloud services list --enabled | - List enabled services |
| gcloud services enable SERVICE_NAME | - Enable a service |
| gcloud logging read "resource.type='gce_instance'" | - Read logs |

SERVICE

PURPOSE

SERVICE	PURPOSE
Compute Engine	Virtual Machines in the cloud
GKE	Run & manage Kubernetes clusters
Cloud Storage	Store objects / files
Cloud SQL	Managed relational databases
Cloud Spanner	Globally distributed SQL DB
BigQuery	Data warehouse for analytics
Cloud CDN	Cache content close to users
Cloud Armor	Protect against DDoS & attacks
Cloud IAM	Manage access & permissions
Cloud Monitoring	Monitor metrics & set alerts
Cloud Logging	Centralized logs management
Cloud Functions	Run event-driven functions
Cloud Run	Run containers serverlessly
VPC	Isolated private cloud network

LAST MINUTE CHECKLIST



- ☑ Understand core services & use cases.
- ☑ Know Regions, Zones & Multi-Region.
- ☑ Revise IAM roles & least privilege.
- ☑ Understand Pricing Models.
- ☑ Know Storage classes & options.
- ☑ Practice gcloud basic commands.
- ☑ Understand Security best practices.
- ☑ Review High Availability & Scalability.
- ☑ Solve use case / scenario based questions.

INTERVIEW / EXAM TIPS



- Start with high level architecture.
- Explain services & their use cases.
- Use diagrams to explain.
- Compare options when asked.
- Focus on Security, HA & Scalability.
- Stay calm & think step by step.

KEY TAKEAWAYS

- GCP offers global, scalable, secure & reliable cloud services.
- Choose the right service for the right use case.
- Focus on core: Compute, Storage, Networking, IAM, Security, Data & Analytics.
- Keep learning & keep building!

REMEMBER



- Think Cloud
- Think GCP
- Practice Daily
- Be Interview Ready ☑

GCP BILLING & COST MANAGEMENT



- Use Budgets to track & control costs.
- Set Budget Alerts to avoid overspending.
- Use Cost Explorer to analyze costs.
- Use Committed Use Discounts (CUDs) for long term savings.
- Use Sustained Use Discounts (SUDs) for automatic savings.
- Use Labels to allocate & track costs.
- Regularly review & delete unused resources.

VPC NETWORKING COMPONENTS



- VPC - Isolated virtual network in GCP.
- Subnet - Range of IP addresses in a region.
- Firewall Rules - Control traffic.
- Routes - Define how traffic is directed.
- Cloud NAT - Allow outbound internet access without external IP.
- VPC Peering - Connect two VPCs privately.
- Shared VPC - Centralized network for multiple projects.

LOAD BALANCING TYPES



- HTTP(S) Load Balancer - For web traffic.
- TCP/SSL Proxy Load Balancer - For TCP/SSL traffic.
- Internal Load Balancer - For internal (private) traffic.
- Network Load Balancer - For high performance TCP/UDP.
- Cloud Load Balancing is global, scalable & highly available.

CLOUD CDN



- Content Delivery Network caches content at edge locations.
- Reduces latency & improves performance.
- Works with HTTP(S) Load Balancer.
- Ideal for static & dynamic content.
- Global anycast IP for best performance.

GCP GLOBAL NETWORK INFRASTRUCTURE

- Regions
- Zones
- Edge Locations



GCP has global infrastructure with multiple regions, zones and edge locations for high availability and low latency.

CLOUD FUNCTIONS



- Serverless compute service.
- Run code in response to events.
- Pay only for the execution time.
- Supports many languages.
- Integrates with GCP services.
- No need to manage servers.

CLOUD RUN vs GKE



	Cloud Run	GKE (Kubernetes Engine)
Type	Serverless	Managed Kubernetes
Management	No infra management	Cluster management
Scaling	Auto scales to zero	Manual / Auto scaling
Use Case	Stateless, event-driven apps, APIs	Complex, microservices, stateful apps
Pricing	Pay per request	Pay for nodes + usage
Flexibility	Less control	Full control

GKE KEY POINTS



- Managed Kubernetes service by Google.
- Automated upgrades & patching.
- Node Pools - Manage groups of nodes.
- Auto Scaling - Scale nodes & pods.
- Workloads - Deploy containers easily.
- Integrates with Cloud Monitoring, Logging, IAM, VPC, etc.

BIGQUERY



- Serverless, highly scalable data warehouse.
- Analyze large datasets with SQL.
- Pay only for data processed.
- Integrates with GCS, Dataflow, Looker.
- Supports real-time analytics.
- Use partitioning & clustering for better performance.

PUB/SUB



- Fully managed real-time messaging service.
- Decouples senders & receivers.
- High throughput & globally scalable.
- At-least-once delivery.
- Use for event-driven architectures.

DATAFLOW



- Fully managed stream & batch data processing.
- Based on Apache Beam.
- Autoscaling workers.
- Integrates with Pub/Sub, BigQuery, GCS, etc.
- Ideal for ETL & real-time pipelines.

FIREBASE (OVERVIEW)



- Platform for mobile & web app development.
- Auth, Firestore, Realtime DB, Storage, Hosting, Functions.
- Easy to start & scale.
- Integrates well with Google Cloud.
- Good for startups & MVPs.

CLOUD ARMOR

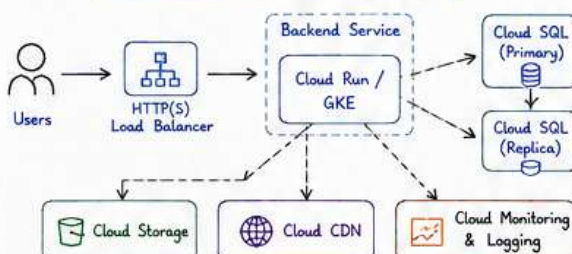


- DDoS protection service.
- Integrates with Cloud Load Balancing & CDN.
- Uses global anycast network.
- Provides WAF (Web Application Firewall).
- Helps protect apps from attacks.

IMPORTANT GCP SERVICES (AT A GLANCE)

Compute Engine	Cloud Storage	BigQuery	Cloud Load Balancing
GKE (Kubernetes Engine)	Cloud SQL	Dataflow	Cloud DNS
Cloud Run	Cloud Spanner	Pub/Sub	Cloud NAT
App Engine	Firestore / Datastore	Looker / Looker Studio	Cloud IAM
Cloud Functions	Bigtable	Vertex AI	Cloud Monitoring
Cloud Batch	Memorystore	Cloud CDN	Cloud Logging

SAMPLE ARCHITECTURE (WEB APP)



EXAM / INTERVIEW TIPS



- ✓ Understand core services & use cases.
- ✓ Know Regions, Zones & Multi-Region.
- ✓ Learn IAM, Roles & least privilege.
- ✓ Understand pricing & cost optimization.
- ✓ Practice architecture diagrams.
- ✓ Focus on security, HA, scalability.
- ✓ Solve use cases & scenario-based questions.
- ✓ Keep hands-on practice.

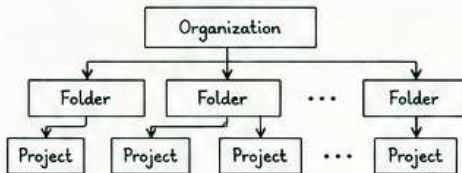
QUICK RECAP



- ☆ GCP provides global, scalable, secure and reliable cloud services.
- ☆ Focus on core services, architecture, security & cost.
- ☆ Practice diagrams & real world use cases.
- ☆ You've Got This! 💪

GCP PROJECTS & ORGANIZATIONS

- Organization is the top level (your company).
- Folder helps to organize projects.
- Project is a collection of resources.
- Each project has unique Project ID.
- Billing is enabled at Project or Folder level.
- Use labels to organize resources.



VM INSTANCE TYPES (Compute Engine)

- General Purpose - Balanced CPU & memory (e2, n2, n2d)
- Compute Optimized - High CPU (c2, c3)
- Memory Optimized - High memory (m2, m3)
- Accelerator Optimized - GPUs / TPUs (a2, a3, t2d)
- Choose machine type based on workload requirement and cost.

GKE ESSENTIALS

- GKE is a fully managed Kubernetes service.
- Control plane is managed by Google.
- You manage worker nodes (or use Autopilot).
- Supports Standard & Autopilot modes.
- Features: Auto-scaling, Auto-upgrade, Self-healing.
- Integrates with IAM, VPC, Cloud Monitoring.

CLOUD STORAGE OPTIONS

Storage Class	Use Case	Min Storage Duration
Standard	Frequently accessed data	-
Nearline	Infrequent access	30 days
Coldline	Rarely accessed data	90 days
Archive	Long-term backup	365 days
Autoclass	Auto moves to appropriate class	-

IAM PRINCIPLES

- Identity - Users, Service Accounts, Groups.
- Principle of Least Privilege - Give only the permissions required.
- Roles contain permissions.
- Predefined roles or custom roles.
- Bindings = (Member + Role + Resource).
- Use IAM Conditions for context-aware access.

CLOUD MONITORING & LOGGING

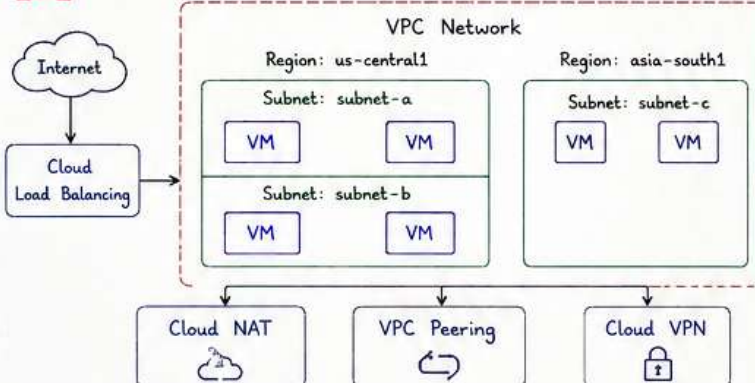
Monitoring (Metrics):

- Collects metrics, provides dashboards & alerts.
- Tracks uptime, CPU, memory, latency, etc.

Logging (Logs Explorer):

- Centralized logs for resources.
- Filter, search, analyze logs.
- Integration with BigQuery for analysis.

VPC NETWORK ARCHITECTURE (Overview)



GCP CORE SERVICES (Quick View)

	Compute Engine	→ Virtual machines in the cloud.
	GKE	→ Managed Kubernetes.
	Cloud Run	→ Run containers serverlessly.
	Cloud Storage	→ Object storage for any data.
	Cloud SQL	→ Managed MySQL, PostgreSQL, SQL Server.
	Cloud Spanner	→ Globally distributed relational DB.
	BigQuery	→ Serverless data warehouse.
	Pub/Sub	→ Real-time messaging service.
	Cloud Functions	→ Serverless functions (event-driven).
	Cloud IAM	→ Manage access and permissions.
	Cloud DNS	→ Managed DNS service.
	Cloud CDN	→ Global content delivery.
	Cloud Armor	→ DDoS protection & WAF.

DATABASES OVERVIEW

Service	Best For
Cloud SQL	Relational DB (MySQL, PostgreSQL, SQL Server) - single region
Cloud Spanner	Globally distributed relational DB (high scale, strong consistency)
Firestore / Datastore	NoSQL document DB (serverless, auto-scaling)
Bigtable	Wide column NoSQL DB (large scale workloads)
Memorystore	Managed Redis / Memcached

MIGRATION OPTIONS

- Migrate for Compute Engine - Lift & shift VMs.
- Migrate for Anthos - Modernize to containers.
- Database Migration Service - Migrate DBs.
- Transfer Appliance - Large data transfer.
- Storage Transfer Service - Transfer data to Cloud Storage.
- BigQuery Data Transfer - Scheduled data import from SaaS / databases.

COST OPTIMIZATION TIPS

- Use Committed Use Discounts (CUDs).
- Use Sustained Use Discounts (SUDs) for VMs.
- Choose right machine type & size.
- Use Preemptible / Spot VMs for batch jobs.
- Auto-scale resources based on demand.
- Delete unused resources & snapshots.
- Use Budget Alerts & Cost Reports.

SECURITY BEST PRACTICES

- Use IAM least privilege access.
- Enable MFA for users.
- Encrypt data at rest and in transit.
- Use VPC Service Controls for data exfiltration protection.
- Regularly audit logs & IAM roles.
- Enable Cloud Armor for DDoS & WAF protection.
- Use Secret Manager for secrets.

DEVOPS & CI/CD

- Cloud Build - Build, test & deploy.
- Artifact Registry - Store container images.
- Cloud Deploy - Deploy to GKE / Run.
- Cloud Source Repositories - Git repos.
- Infrastructure as Code - Use Terraform / Deployment Manager.
- Cloud Monitoring - Monitor applications.

IMPORTANT GCP COMMANDS (gcloud)

- gcloud auth login
- gcloud config set project PROJECT_ID - Set project
- gcloud projects list - List all projects
- gcloud compute instances list - List VM instances
- gcloud config list - Show configurations
- gcloud iam roles list - List IAM roles
- gcloud services list --enabled - List enabled services
- gcloud components update - Update gcloud
- gcloud logging read "resource.type=gce_instance" - Read logs

INTERVIEW TIPS

- Understand core services & use cases.
- Know Regions, Zones & Multi-Region.
- Practice architecture diagrams.
- Be clear on IAM, Security & Networking.
- Compare services (e.g., Cloud Run vs GKE, Cloud SQL vs Spanner).
- Focus on HA, Scalability & Cost.
- Practice hands-on in Console / CLI.
- Explain trade-offs clearly.

GCP SERVICE ACCOUNTS



- Service Accounts represent your application on GCP.
- Have IAM roles, keys & permissions.
- Used by apps / VMs to access GCP resources securely.
- Best Practice: Use least privilege and rotate keys.

NETWORK SECURITY TOOLS



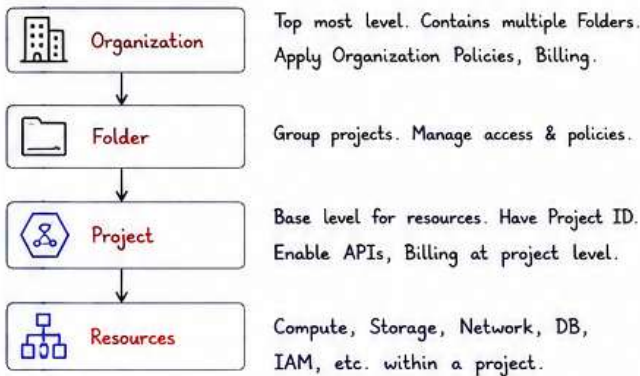
- VPC Firewall Rules - Allow / deny traffic.
- Cloud Armor - DDoS & WAF protection.
- VPC Service Controls - Protect data exfiltration.
- Identity-Aware Proxy (IAP) - Secure access to apps without VPN.
- SSL/TLS Certificates - Managed by Certificate Manager.

GCP PRICING MODELS

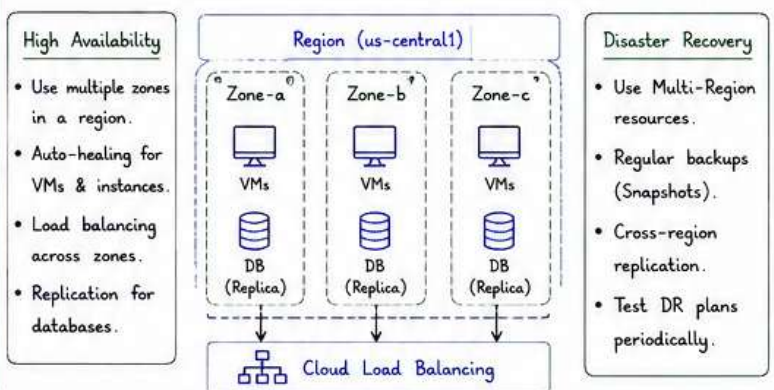


- **Pay-as-you-go** - Pay for what you use.
- **Committed Use Discounts** - 1 or 3 years commitment for discount.
- **Sustained Use Discounts** - Automatic discount for consistent usage.
- **Preemptible / Spot VMs** - Up to 80% cheaper but can be interrupted.
- **Free Tier** - Limited free resources.

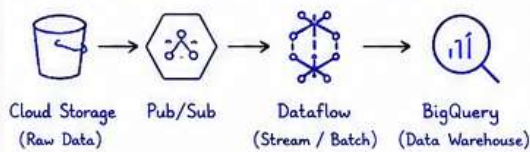
GCP RESOURCE HIERARCHY (Detailed View)



HIGH AVAILABILITY & DISASTER RECOVERY



DATA FLOW PIPELINE (Example)

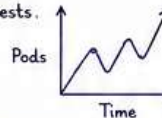


Use Case: Real-time analytics, ETL pipelines, IoT data processing, log analysis.

GKE AUTO SCALING



- Cluster Autoscaler - Automatically adjusts number of nodes.
- Horizontal Pod Autoscaler (HPA) - Adjusts pods based on CPU / memory.
- Vertical Pod Autoscaler (VPA) - Adjusts pod resource requests.
- Ensures performance & cost optimization.



TYPES OF GCP NETWORKS



- VPC (Virtual Private Cloud) - Isolated virtual network.
- Shared VPC - Centralized VPC for multiple projects.
- VPC Peering - Connect two VPCs privately.
- Cloud VPN - Secure connectivity over internet.
- Dedicated Interconnect - Private connection via Google backbone.

GCP SERVICES BY CATEGORY

Category	Services
Compute	Compute Engine, GKE, Cloud Run, App Engine, Cloud Functions, Cloud Batch
Storage	Cloud Storage, Persistent Disk, Filestore, Cloud SQL Storage
Database	Cloud SQL, Cloud Spanner, Firestore, Bigtable, Memorystore, AlloyDB, Datastore
Networking	VPC, Cloud Load Balancing, Cloud CDN, Cloud DNS, Cloud NAT, Cloud VPN, Interconnect
Data & Analytics	BigQuery, Dataflow, Dataproc, Pub/Sub, Looker, Vertex AI
Security	IAM, Cloud Armor, Security Command Center, Secret Manager, KMS
Management	Cloud Monitoring, Logging, Error Reporting, Deployment Manager, Operations Suite

IMPORTANT GCP CONCEPTS



- ★ Project - Logical grouping of resources.
- ★ Region - Geographic location.
- ★ Zone - Isolated location within a region.
- ★ Resource - Any service or component.
- ★ IAM - Access and Identity Management.
- ★ API - Interface to interact with services.
- ★ Quota - Limits for resources/services.
- ★ Labels - Key-Value pairs for organization.
- ★ Metadata - Data about resources.
- ★ SLA - Service Level Agreement.

LAST MINUTE CHECKLIST



- ☒ Understand core services & use cases.
- ☒ Know Regions, Zones & Multi-Region.
- ☒ Learn IAM roles & best practices.
- ☒ Understand pricing & cost optimization.
- ☒ Practice architecture diagrams.
- ☒ Focus on Security, HA & Scalability.
- ☒ Solve use cases & scenario questions.
- ☒ Keep hands-on practice in Console.
- ☒ Revise important gcloud commands.
- ☒ Stay calm & confident 😊

COMMON GCP USE CASES



- Host scalable web apps → App Engine / Cloud Run
- Run microservices → GKE / Cloud Run
- Store files & backups → Cloud Storage
- Run databases → Cloud SQL / Spanner
- Real-time analytics → BigQuery / Dataflow
- Machine learning → Vertex AI
- CI/CD Pipelines → Cloud Build
- Monitor & alert → Cloud Monitoring
- Secure apps → IAM, Cloud Armor, KMS

GCP FREE TIER HIGHLIGHTS



- \$300 free credits for 90 days.
 - Always free products (limited usage):
 - Compute Engine (1 f1-micro / month)
 - Cloud Storage (5 GB standard)
 - Cloud SQL (SQLite, MySQL small)
 - BigQuery (1 TB queries / month)
 - Cloud Functions (2M invocations / month)
 - Cloud Run (2M requests / month)
- Note: Check latest free tier details on GCP website.

QUICK RECAP



GCP provides powerful, scalable, secure and cost-effective cloud solutions. Master core services, understand architecture & keep practicing!

You've Got This! 🍀

COMPUTE ENGINE (DETAILED)



- Infrastructure as a Service (IaaS).
- Create & run VMs in Google's infrastructure.
- Machine Types: General purpose, Compute optimized, Memory optimized, GPU.
- Disks: Standard Persistence (HDD), SSD Persistent Disk, Local SSD.
- Images: Public images, Custom images, Snapshots.
- Features: Live migration, Sole-tenant Nodes, Per-second billing for some VMs.

KUBERNETES ENGINE (GKE) DEEP DIVE



- Managed environment to run Kubernetes clusters.
- Cluster modes: Standard, Autopilot.
- Features: Auto-scaling, Self-healing, Rolling updates, Auto-upgrade.
- Networking: VPC-native, Alias IP ranges.
- Security: Workload Identity, Binary Authorization.
- Storage: Persistent Volumes (PD, SSD PD, Filestore, etc.).
- Best for containerized, microservices based apps.

CLOUD IAM ROLES (EXAMPLES)



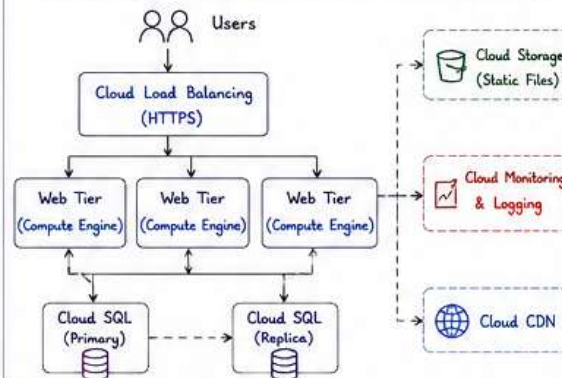
- Owner - Full access to all resources.
 - Editor - Modify resources.
 - Viewer - Read-only access.
 - Compute Admin - Manage Compute resources.
 - Storage Admin - Manage Storage resources.
 - Security Admin - Manage security policies.
 - Custom Roles - Create custom permissions.
- Tip: Grant least privilege.

VPC NETWORKING (ADVANCED)



- Global VPC - One network across all regions.
- Subnet Modes: Auto mode, Custom mode.
- Firewall Rules: Allow/Deny traffic (Ingress/Egress).
- Routes: Static routes, Dynamic routing (Cloud Router).
- NAT: Cloud NAT allows outbound internet without external IPs.
- VPC Peering: Private connectivity between VPCs.
- Shared VPC: Centralized network for many projects.
- VPC Service Controls: Secure service perimeters.

CLOUD ARCHITECTURE PATTERN - 3 TIER WEB APP



CLOUD STORAGE (USE CASES)



- Store unstructured data - images, videos, backups, logs.
- Storage Classes:
 - Standard - Frequent access
 - Nearline - Infrequent access
 - Coldline - Rare access
 - Archive - Long term backup
- Features: Versioning, Lifecycle management, IAM access.
- Use with CDN for faster delivery of static content.

CLOUD SQL VS FIRESTORE

	Cloud SQL	Firestore
Type	Relational DB (SQL)	NoSQL Document DB
Model	Tables, Rows	Documents, Collections
Scalability	Vertical / Manual	Horizontal / Automatic
Consistency	Strong (ACID)	Eventual
Best For	Structured data, complex queries	Realtime apps, Mobile, IoT, Rapid dev
Examples	MySQL, PostgreSQL, SQL Server	Firestore (NoSQL)

BIGQUERY - QUICK NOTES



- Serverless, highly scalable data warehouse.
- Use Standard SQL.
- Pay only for data scanned.
- Integrates with GCS, Dataflow, Looker.
- Features: Partitioned tables, Clustering, Materialized views.
- Best for analytics, dashboards, large data processing.

CLOUD FUNCTIONS



- Serverless - run code without managing servers.
- Triggered by events (HTTP, Storage, Pub/Sub, Firestore, etc.).
- Pay only for execution time.
- Supports many runtimes (Node.js, Python, Go, Java, etc.).
- Ideal for microservices, automation, background tasks.

PUB/SUB



- Fully managed real-time messaging service.
- Publishers send messages to a topic.
- Subscribers receive messages.
- At-least-once delivery.
- Great for decoupling services.
- Use cases: Event-driven architecture, streaming data.

DATAFLOW



- Stream and batch data processing service.
- Built on Apache Beam.
- Auto-scaling workers.
- Integrates with Pub/Sub, BigQuery, GCS.
- Use for ETL, real-time pipelines.
- Supports exactly-once processing.

FIRESTORE (IN DEPTH)



- NoSQL document database.
- Data stored in documents (JSON-like).
- Real-time synchronization.
- Offline support for mobile/web.
- Automatic scaling & sharding.
- Strong security with IAM & Rules.
- Use for mobile apps, chat apps, IoT, CMS.

EDGE SERVICES



- Cloud CDN - Global content delivery.
- Cloud Armor - DDoS & WAF protection.
- Cloud Load Balancing - Global traffic distribution.
- Cloud DNS - Managed DNS service.
- These improve performance, security & reliability.

DEVOPS & CI/CD ON GCP



- Cloud Build - Build, test, deploy.
- Artifact Registry - Store container images.
- Cloud Deploy - Deploy to GKE / VMs.
- Cloud Source Repositories - Git repos.
- Infrastructure as Code - Terraform, Deployment Manager.
- Cloud Monitoring - Monitor applications.
- Cloud Logging - Centralized logs.

IMPORTANT GCP SHORTCUT COMMANDS (gcloud)



```

gcloud auth login           - Login to GCP
gcloud config set project PROJECT_ID - Set project
gcloud projects list       - List all projects
gcloud compute instances list - List VM instances
gcloud compute instances create NAME --zone=ZONE - Create VM
gcloud compute instances delete NAME --zone=ZONE - Delete VM
gcloud storage buckets list - List all buckets
gcloud config list         - Show configurations
gcloud services list --enabled - List enabled services
gcloud logging read "resource.type=gce_instance" Read logs

```

HIGH AVAILABILITY TIPS



- Use Multiple Zones / Regions.
- Use Load Balancing across zones.
- Use Managed services (GKE, Cloud SQL).
- Enable Auto-healing & Auto-scaling.
- Use Persistent Disks (not Local SSD).
- Replicate data across regions.

EXAM / INTERVIEW QUICK TIPS



- ✓ Understand core services & use cases.
- ✓ Know Regions, Zones, Multi-Region.
- ✓ Practice IAM, IAM roles, permissions.
- ✓ Focus on Security, HA & Scalability.
- ✓ Draw architecture diagrams.
- ✓ Solve scenario-based questions.
- ✓ Keep learning & stay updated!

REMEMBER



- Plan well before deploying.
- Start small, scale as you grow.
- Monitor cost, performance & security.
- Think Cloud ☁, Think GCP 🚀
- You've Got This! 💪

GCP REGIONS, ZONES & MULTI-REGION

- **Region** - A large geographic location (e.g., us-central1, asia-south1)
- **Zone** - Isolated location within a region (e.g., us-central1-a, us-central1-b)
- **Multi-Region** - Spans multiple regions (e.g., US, EU, ASIA)
- Use multiple zones for High Availability.
- Use Multi-Region for global services (e.g., Cloud Storage, BigQuery).

GCP PRICING & BILLING

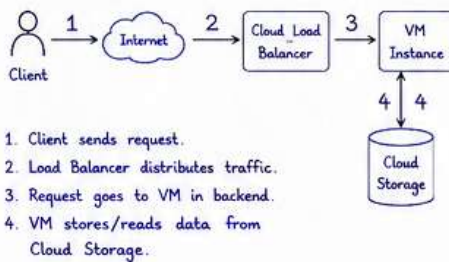
- Pay-as-you-go - No upfront commitment.
- Committed Use Discounts - 1 or 3 years.
- Sustained Use Discounts - Automatic discounts for consistent usage.
- Preemptible / Spot VMs - Cheaper but can be interrupted.
- Budgets & Alerts - Track and control costs.
- Billing Reports - Analyze cost by project, service, label, etc.

GCP SUPPORT PLANS

Plan	Best For	Features
Basic	Getting started	Community support, docs, billing support
Standard	Production workloads	Business hours support, response in 24 hrs
Enhanced	Mission critical apps	24x7 support, 1 hr response time.
Premium	Business critical apps	24x7 support, 15 min response time, TAM

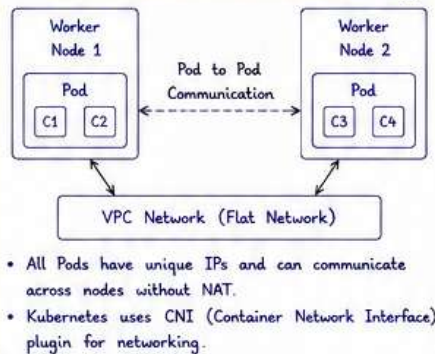
Tip: Choose support plan based on business needs.

VPC FLOW - HOW IT WORKS



1. Client sends request.
2. Load Balancer distributes traffic.
3. Request goes to VM in backend.
4. VM stores/reads data from Cloud Storage.

KUBERNETES POD NETWORKING

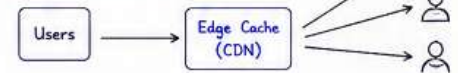


- All Pods have unique IPs and can communicate across nodes without NAT.
- Kubernetes uses CNI (Container Network Interface) plugin for networking.

CLOUD CDN - OVERVIEW

- Content Delivery Network caches content at edge locations.
- Delivers content closer to users.
- Reduces latency & improves performance.
- Works with HTTP(S) Load Balancer.
- Ideal for static & dynamic content.
- Supports global anycast IP.

Use Cases: Websites, media streaming, software downloads, APIs.



BIGQUERY - IN DEPTH

- Serverless, highly scalable data warehouse.
- Stores data in columnar format.
- Uses Standard SQL.
- Charges based on data scanned.
- Integrates with Dataflow, Cloud Storage, Looker, Pub/Sub.
- Features: Partitioned tables, Clustering, Materialized views.

Use Cases: Analytics, reporting, dashboards, large scale data analysis.

CLOUD SQL - IN DEPTH

- Managed relational database service.
- Supports MySQL, PostgreSQL, SQL Server.
- Automated backups, patching, failover.
- Read Replicas for scalability.
- Private IP connectivity through VPC.
- High Availability with Regional instances.

Use Cases: Web apps, ERP, CRM, transactional applications.

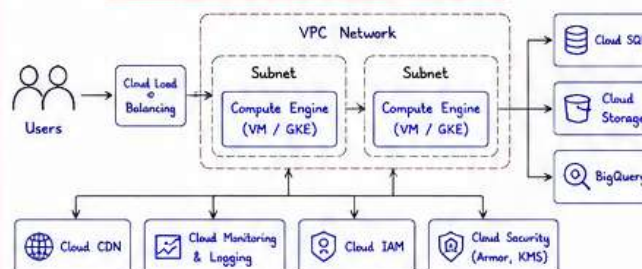
CLOUD SPANNER - IN DEPTH

- Globally distributed relational database.
- Horizontal scaling with strong consistency.
- 99.999% availability.
- Supports SQL.
- Use Cases: Banking, retail, gaming, IoT applications.
- Designed for mission-critical workloads.

IAM ROLES - COMMON ROLES

Role	Description
Viewer	Read-only access to see resources.
Editor	Modify existing resources.
Owner	Full control over all resources.
Compute Admin	Manage Compute Engine resources.
Storage Admin	Manage Cloud Storage resources.
Security Admin	Manage IAM, security policies.
Billing Admin	Manage billing & cost settings.

GCP ARCHITECTURE - HIGH LEVEL VIEW



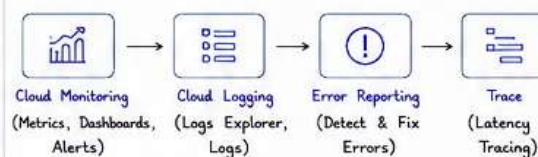
FIREBASE - OVERVIEW

- Platform for mobile & web app dev.
- Auth, Firestore, Realtime DB, Storage, Hosting, Functions.
- Easy to build & scale apps.
- Integrates well with GCP services.
- Use Cases: Chat apps, real-time apps, MVPs, startups.

COMPUTE OPTIONS COMPARISON

Service	Best For	Management
Compute Engine	IaaS VMs	You manage VMs
GKE	Containers	Google manages K8s
Cloud Run	Containers / Apps	Fully managed
App Engine	PaaS Apps	Fully managed
Cloud Functions	Event driven	Fully managed

MONITORING & OBSERVABILITY STACK



SECURITY TOOLS OVERVIEW

- Cloud Armor - DDoS protection & WAF.
- Identity-Aware Proxy - Secure access.
- Security Command Center - Risk & posture management.
- Binary Authorization - Secure container images.
- KMS - Manage encryption keys.
- Secret Manager - Manage secrets.

DATA & ANALYTICS SERVICES

- BigQuery - Data warehouse.
- Dataflow - Stream & batch processing.
- Dataproc - Managed Hadoop / Spark.
- Pub/Sub - Real-time messaging.
- Looker / Looker Studio - Dashboards.
- Vertex AI - ML platform for models.

IMPORTANT TIPS

- ✓ Understand core services & their use cases.
- ✓ Know Regions, Zones & Multi-Region.
- ✓ Learn IAM, networking & security.
- ✓ Practice architecture diagrams.
- ✓ Focus on cost optimization.
- ✓ Solve real world & scenario based questions.
- ✓ Keep hands-on practice in Console / CLI.

GCP FREE TIER - QUICK LOOK

- \$300 credits for first 90 days.
- Always free products (limited usage):
 - Compute Engine (1 f1-micro / month)
 - Cloud Storage (5 GB standard)
 - Cloud SQL (SQLite, MySQL small)
 - BigQuery (1 TB queries / month)
 - Cloud Functions (2M invocations / month)

Note: Check latest details on GCP website.

QUICK RECAP

Think Cloud ☁, Think GCP 🚀
Learn Today,
Lead Tomorrow!
You've Got This! 💪

LOAD BALANCING - DEEP DIVE



Types:

- Global HTTP(S) Load Balancer (L7)
- Global TCP/SSL Proxy (L4)
- Regional Internal Load Balancer
- Network Load Balancer (L4, high perf)

Features:

- Global anycast IP
- Health checks
- Auto-scaling with backend services
- SSL termination
- Cloud CDN integration
- Works across multiple regions

CLOUD DNS - OVERVIEW



- Fully managed, scalable, authoritative DNS service.
- Low latency, high availability.
- Supports public and private zones.
- Integrates with VPC and GKE.
- Features: DNSSEC, query logging, geo-location routing, traffic steering.
- Use Cases: Domain management, service discovery, multi-region routing.

CLOUD ARMOR - SECURITY



- Managed DDoS protection and WAF.
- Protects against Layer 3, 4 and 7 attacks.
- Integrates with Cloud Load Balancing.
- Features:
 - DDoS mitigation (always on)
 - Web Application Firewall (OWASP rules)
 - Custom security policies
 - Geo-blocking, IP allow/deny
 - Rate limiting (throttling)
- Use Case: Secure web applications from bots, SQL injection, XSS, etc.

CLOUD INTERCONNECT vs VPN

Cloud VPN

- Encrypted connection over the internet.
- Lower cost.
- Easy to set up.
- Good for small workloads.
- Uses IPsec.



Cloud Interconnect

- Dedicated private connection to Google.
- Higher performance and reliability.
- Lower latency.
- Used for large / production workloads.



ANTHOS - HYBRID & MULTI-CLOUD



- Platform to run and manage applications across on-prem, multi-cloud and GCP.
- Supports Kubernetes (GKE).
- Provides consistent deployment, security and policies.
- Features: Anthos Config Management, Service Mesh, Policy Controller, Fleet Management.
- Use Case: Hybrid cloud, multi-cloud strategy, modernizing legacy apps.

VERTEX AI - OVERVIEW



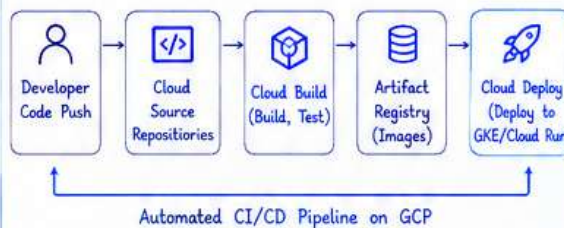
- Unified platform for ML on Google Cloud.
- Build, train, deploy and manage ML models.
- Supports AutoML and custom training.
- Integrates with BigQuery, Dataflow, etc.
- Features:
 - Notebooks, Training, Model Registry
 - Managed Endpoints for predictions
 - Pipelines for MLOps
- Use Cases: Predictive analytics, NLP, computer vision, recommendation systems.

GCP SERVICE INTEGRATIONS



- Services work seamlessly together.
- Examples:
 - Cloud Run + Cloud Build (CI/CD)
 - GKE + Cloud Monitoring + Logging
 - BigQuery + Dataflow + Looker
 - Pub/Sub + Cloud Functions
 - Cloud Storage + Vertex AI
- Benefit: Build end-to-end scalable solutions with minimal configuration.

SAMPLE CI/CD PIPELINE (GCP)



MONITORING METRICS (IMPORTANT)



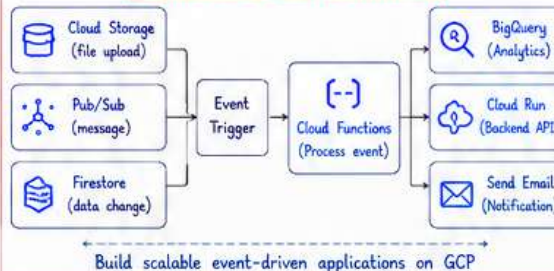
- Monitored Metrics:
 - CPU, Memory, Disk, Network
 - Request count, latency, error rate
- Tools:
 - Cloud Monitoring (Metrics, Dashboards)
 - Cloud Logging (Logs Explorer)
 - Error Reporting (Track errors)
 - Trace (Distributed tracing)
- Set up Alerts for thresholds (e.g., high CPU, high error rate).

CLOUD RUN - USE CASES



- Run stateless containers without managing servers.
- Auto-scales to zero.
- Pay only for execution time.
- Supports HTTP services, background jobs, event-driven workloads.
- Use Cases: APIs, microservices, web apps, scheduled jobs, lightweight services.

ARCHITECTURE: EVENT-DRIVEN APP



COST OPTIMIZATION STRATEGIES



- Use Committed Use Discounts (1 or 3 years).
- Use Sustained Use Discounts (automatic).
- Choose right machine types and sizes.
- Use Preemptible / Spot VMs for batch jobs.
- Auto-scale resources based on demand.
- Delete unused resources (VMs, disks, IPs).
- Use Budget Alerts and Cost Reports.
- Optimize storage classes and lifecycle rules.
- Regularly review and right-size.

GCP vs AWS vs AZURE (QUICK)



Feature	GCP	AWS	Azure
Compute	Compute Engine, GKE	EC2, EKS	VMs, AKS
Serverless	Cloud Run, Functions	Lambda	Functions
Storage	Cloud Storage	S3	Blob Storage
Data Warehouse	BigQuery	Redshift	Synapse
Kubernetes	GKE	EKS	AKS
Unique Strength	Data & AI	Wide services	Enterprise integration

REAL WORLD USE CASES



- Streaming Data Pipeline: Pub/Sub → Dataflow → BigQuery → Looker.
- E-commerce Platform: Cloud Run + Cloud SQL + Cloud CDN + Cloud Armor.
- ML Platform: Vertex AI + BigQuery.
- Global Web App: Cloud Load Balancing + GKE (multi-region).
- Hybrid Cloud: Anthos.
- Media Streaming: Cloud Storage + CDN.
- IoT: Pub/Sub + Dataflow + BigQuery.

FINAL REVISION CHECKLIST



- ☒ Core services & use cases
- ☒ Regions, Zones & Multi-Region
- ☒ IAM roles & security best practices
- ☒ Networking (VPC, Subnet, Firewall, Load Balancer)
- ☒ Storage options and use cases
- ☒ Databases (Cloud SQL, Spanner, Firestore, BigQuery)
- ☒ Compute options (Compute Engine, GKE, Cloud Run)
- ☒ Serverless, Dataflow, Pub/Sub
- ☒ Monitoring, Logging, Error Reporting
- ☒ Cost optimization strategies
- ☒ Architecture diagrams practice
- ☒ Hands-on practice in Console
- ☒ Be ready for scenario-based questions

Consistent
Learning
Creates
Big Opportunities
! ❤️

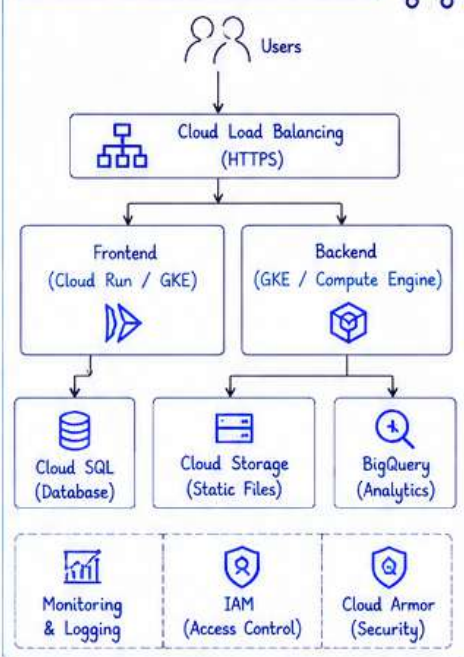
GCP INTERVIEW Q&A (QUICK REVISION)

- Q1. What is GCP?
→ A cloud platform by Google to build, deploy and scale applications.
- Q2. Difference between Project, Folder, Organization?
→ Org (top) > Folder (group) > Project (resources).
- Q3. What is the default network in GCP?
→ Auto mode VPC with predefined subnets.
- Q4. How does GCP ensure security?
→ IAM, VPC, Cloud Armor, encryption, security command center.
- Q5. What is the difference between Compute Engine and Cloud Run?
→ Compute Engine = VMs (IaaS), Cloud Run = serverless containers (PaaS).
- Q6. When to use BigQuery?
→ For large scale analytics and data warehousing.
- Q7. How to reduce cost in GCP?
→ Use committed use discounts, right-size VMs, preemptible VMs, delete unused resources.

SERVICE SELECTION GUIDE

Use Case	Recommended Service
Run VMs	Compute Engine
Run containers (managed)	GKE
Run serverless containers	Cloud Run
Build & deploy apps	App Engine
Store structured data	Cloud SQL
Store unstructured data	Cloud Storage
Large scale analytics	BigQuery
Stream processing	Dataflow / Pub/Sub
Machine Learning	Vertex AI
Static website hosting	Cloud Storage + CDN
DNS management	Cloud DNS
Networking (isolated)	VPC
Security (DDoS, WAF)	Cloud Armor
Monitoring & Logging	Cloud Monitoring + Logging

COMMON ARCHITECTURE DIAGRAM



TROUBLESHOOTING TIPS

- Check Cloud Logging for error details.
- Use Cloud Monitoring dashboards.
- Verify IAM permissions (permission denied errors).
- Check VPC firewall rules (connectivity issues).
- Use gcloud CLI to test configurations.
- Validate service health status (Google Cloud Status Dashboard).
- For slow performance, check latency, region, and resource utilization.
- Common fixes: restart instances, update firewall rules, check quotas.

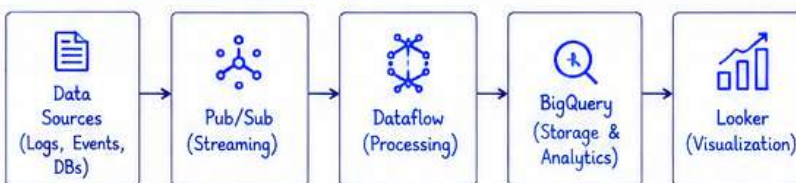
QUOTAS & LIMITS

- Each project has default quotas (e.g., CPUs, storage, APIs).
- Can request quota increase from GCP Console.
- Common quotas: Compute Engine instances, Cloud Storage buckets, BigQuery queries.
- Monitor quota usage to avoid service disruption.
- Use multiple projects for large workloads.

USEFUL GCP LINKS

- GCP Console: <https://console.cloud.google.com>
- Documentation: <https://cloud.google.com/docs>
- Free Tier: <https://cloud.google.com/free>
- Pricing Calculator: <https://cloud.google.com/products/calculator>
- Status Dashboard: <https://status.cloud.google.com>
- gcloud CLI Docs: <https://cloud.google.com/sdk/gcloud>
- Training & Certification: <https://cloud.google.com/training>
- Qwiklabs (Hands-on): <https://www.qwiklabs.com>

DATA PIPELINE ARCHITECTURE



DEPLOYMENT STEPS (CLOUD RUN EXAMPLE)

- Write application code (e.g., Dockerfile).
- Build container image (Cloud Build or locally).
- Push image to Artifact Registry.
- Deploy to Cloud Run (gcloud run deploy).
- Configure environment variables, IAM.
- Expose via HTTPS (auto-managed).
- Monitor logs and performance.

KEY COMPARISONS

Feature	Compute Engine	Cloud Run	GKE	App Engine
Type	IaaS (VMs)	PaaS (Serverless)	Managed K8s	PaaS
Manage Servers	Yes	No	No	No
Scaling	Manual / Auto	Automatic	Automatic	Automatic
Use Case	Full control	Event-driven apps	Containers at scale	Web apps (PaaS)
Best For	Custom VMs	Microservices	Complex apps	Simple apps
Cost	Pay for VMs	Pay per request	Pay for nodes	Pay per usage

CERTIFICATIONS (POPULAR)

- Associate Cloud Engineer (ACE)
- Professional Cloud Architect (PCA)
- Professional Data Engineer
- Professional Cloud Developer
- Professional Cloud DevOps Engineer
- Professional Cloud Security Engineer
- Professional Machine Learning Engineer

💡 Tip: Hands-on practice in Qwiklabs helps a lot for certifications.

FINAL TAKEAWAYS

- ✓ Understand core services & use cases.
- ✓ Learn networking, IAM, and security.
- ✓ Practice architecture diagrams.
- ✓ Focus on real-world use cases.
- ✓ Get hands-on with GCP console.
- ✓ Follow cost optimization best practices.
- ✓ Stay updated with new launches.
- ✓ Practice scenario-based questions.
- ✓ Build small projects to gain confidence.
- ✓ Think Cloud, Think GCP





1. HIGH TRAFFIC WEB APPLICATION

Q: You have a web application which receives sudden high traffic. How will you design it on GCP to handle it?

A:

- Use Cloud Load Balancing (global HTTP(S)).
- Deploy application on GKE or Compute Engine with Auto-scaling.
- Use Cloud CDN for static content.
- Use Cloud Monitoring to set alerts.
- Use Cloud Armor to protect from DDoS.
- Use multi-region deployment for high availability.

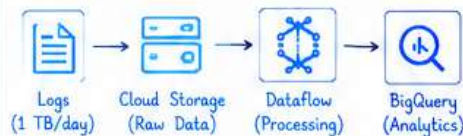


2. STORE AND ANALYZE LARGE DATA

Q: You have 1 TB of log data per day. How will you store and analyze it in GCP?

A:

- Store raw logs in Cloud Storage (cheap, durable).
- Use Dataflow for real-time / batch processing.
- Store processed data in BigQuery.
- Use Looker or Looker Studio for visualization.
- Set lifecycle rules to move old data to Nearline / Coldline storage.



3. MIGRATE ON-PREM TO GCP

Q: Your company wants to migrate on-prem applications to GCP. How will you approach it?

A:

- 1 Assess current infrastructure and dependencies.
- 2 Choose migration strategy (Lift & Shift, Re-platform, Re-architect).
- 3 Use Migrate for Compute Engine / Database Migration Service.
- 4 Set up hybrid connectivity using Cloud VPN or Interconnect.
- 5 Test in staging environment.
- 6 Migrate in phases with minimal downtime.
- 7 Monitor and optimize after migration.



4. DESIGN A SECURE ARCHITECTURE



Q: How will you design a secure architecture on GCP?

A:

- Use IAM with least privilege.
- Enable VPC with private subnets.
- Use Cloud Armor for DDoS & WAF protection.
- Encrypt data using CMEK / KMS.
- Use Secret Manager for sensitive credentials.
- Enable VPC Service Controls.
- Monitor with Security Command Center.
- Follow organization policies and audit logs.

5. CONNECT TWO VPCs



Q: How will you connect two VPC networks in GCP?

A:

- Use VPC Peering for private connectivity (low latency, no external IP needed).
- Ensure non-overlapping IP ranges.
- Alternatively, use Cloud VPN or Cloud Interconnect for hybrid connectivity.



6. DATABASE CHOICES



Q: Which GCP database service will you use for different use cases?

A:

- Cloud SQL → Relational DB (MySQL, Postgres, SQL Server) for transactional apps.
- Firestore → NoSQL document database (serverless).
- Bigtable → Wide column DB for large scale (analytics, IoT).
- Spanner → Globally distributed relational DB for mission-critical apps.
- Memorystore → In-memory cache (Redis).

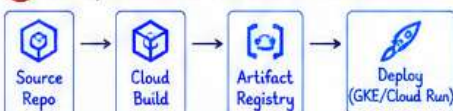
7. IMPLEMENT CI/CD PIPELINE



Q: How will you set up CI/CD for a containerized application on GCP?

A:

- 1 Store code in Cloud Source Repositories.
- 2 Use Cloud Build to build Docker image.
- 3 Push image to Artifact Registry.
- 4 Deploy to GKE / Cloud Run using Cloud Deploy.
- 5 Use Cloud Monitoring for deployment health.
- 6 Set up automated rollbacks if needed.



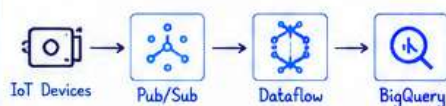
8. REAL-TIME DATA PROCESSING



Q: How will you process real-time streaming data (e.g., IoT, clickstream)?

A:

- Use Pub/Sub for ingesting streaming data.
- Use Dataflow (Apache Beam) for stream processing.
- Store results in BigQuery or Cloud Storage.
- Use Looker for real-time dashboards.



9. COST OPTIMIZATION SCENARIO



Q: Your monthly GCP bill is increasing. How will you reduce costs?

A:

- Use committed use discounts (1 or 3 years).
- Use preemptible/spot VMs for non-critical workloads.
- Delete unused resources (disks, IPs, snapshots).
- Right-size machine types.
- Use Cloud Storage lifecycle rules (Nearline / Coldline).
- Optimize BigQuery queries (partitioning, clustering).
- Use cost management tools and budget alerts.

10. DISASTER RECOVERY PLAN



Q: How will you design a disaster recovery solution on GCP?

A:

- Use multi-region deployment.
- Use Cloud Storage with cross-region replication.
- Enable automated backups (Cloud SQL, Firestore).
- Use load balancing with failover.
- Test DR plan periodically.
- Define RTO (Recovery Time Objective) and RPO (Recovery Point Objective).

11. CHOOSE RIGHT SERVICE



Q: You need to run a simple REST API with low maintenance. Which service will you use?

A:

- Use Cloud Run (serverless containers).
- Auto-scales to zero.
- Pay only for actual usage.
- No server management.
- Good for small to medium workloads.

12. MONITORING & ALERTING



Q: How will you set up monitoring and alerting for your application?

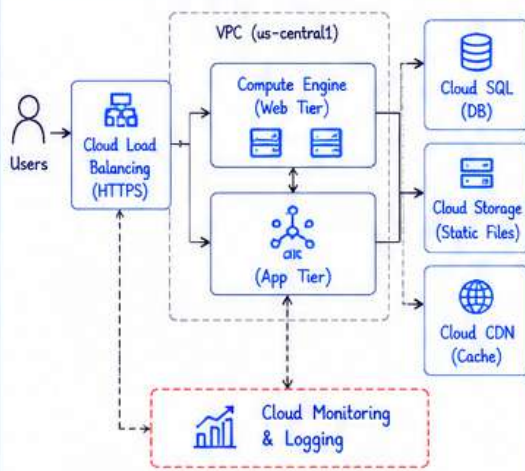
A:

- Use Cloud Monitoring (metrics, dashboards).
- Use Cloud Logging (logs explorer).
- Set up alerting policies (email / Slack / PagerDuty).
- Monitor key metrics: CPU, memory, latency, error rate.
- Use Error Reporting to track application errors.
- Create uptime checks for external endpoints.

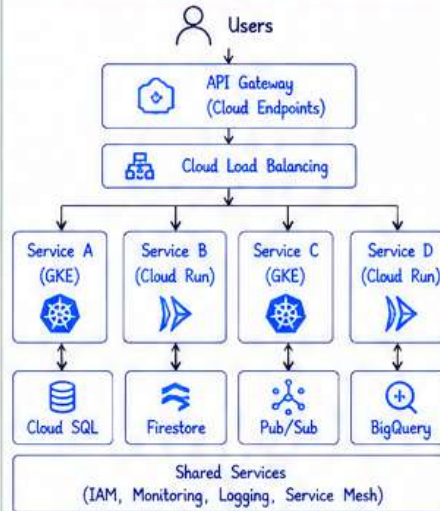
ARCHITECTURE PATTERNS, TROUBLESHOOTING & BEST PRACTICES

Design
Debug
Optimize
Like a Pro!

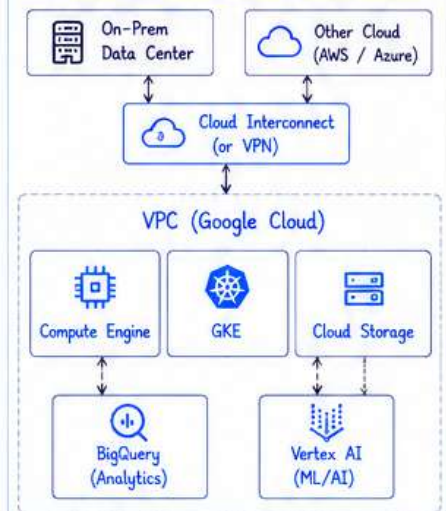
1. REFERENCE ARCHITECTURE - 3 TIER APP



2. MICROSERVICES ARCHITECTURE ON GCP



3. HYBRID & MULTI-CLOUD ARCHITECTURE



4. COMMON GCP ARCHITECTURE PATTERNS

Pattern	Use Case	GCP Services
3-Tier Architecture	Web applications	Load Balancer, GCE, Cloud SQL, CDN
Microservices	Scalable applications	GKE, Cloud Run, Cloud Endpoints
Event-Driven	Asynchronous processing	Pub/Sub, Cloud Functions, Dataflow
Data Lake	Large scale analytics	Cloud Storage, BigQuery, Dataproc
Hybrid Cloud	Connect on-prem with GCP	Cloud Interconnect, VPN
Multi-Region	High availability	Global Load Balancer, Multi-region storage
Serverless	No server management	Cloud Run, Cloud Functions, Firebase
ML/AI Pipeline	Model training & serving	Vertex AI, BigQuery, Cloud Storage
Real-time Streaming	IoT, logs, clickstream	Pub/Sub, Dataflow, BigQuery

5. TROUBLESHOOTING GUIDE

Issue	Possible Cause	Solution
Cannot connect to VM	Firewall rule, wrong IP, no external IP	Check VPC firewall rules, verify IP, use IAP/SSH
High latency in application	Wrong region, no CDN, large data	Use nearest region, enable Cloud CDN, optimize queries
Permission denied error	IAM role missing	Check IAM policy, grant required role (minimum privilege)
Service unavailable	Instance down, health check failing	Check instance health, verify load balancer health checks
Slow BigQuery queries	No partitioning, scanning large data	Use partitioned tables, cluster data, optimize query
Data not showing in logs	Incorrect log filter	Check log name, resource type, and time range
Deployment failure	Wrong config, quota issue	Check Cloud Build logs, verify quotas, fix config

6. DEBUGGING CHECKLIST

- ☒ Check project ID and correct region/zone.
- ☒ Verify IAM permissions.
- ☒ Check VPC firewall rules.
- ☒ Validate service account keys (if using).
- ☒ Review Cloud Logging for error details.
- ☒ Check service health in Cloud Console.
- ☒ Verify environment variables and config.
- ☒ Test connectivity (ping, curl, IAP).
- ☒ Check quotas and limits.
- ☒ Use Cloud Monitoring metrics and alerts.
- ☒ Reproduce issue in a smaller environment.
- ☒ Engage Google Cloud Support if needed.

7. PERFORMANCE OPTIMIZATION TIPS

- Use right machine types (e2, n2, etc.).
- Enable autoscaling (GKE / Instance Groups).
- Use Cloud CDN for static content.
- Use caching (Memorystore - Redis).
- Optimize database queries (BigQuery / Cloud SQL).
- Use committed use discounts / savings plans.
- Delete unused resources.
- Monitor with Cloud Monitoring and set alerts.

8. HIGH AVAILABILITY & FAULT TOLERANCE

- Deploy across multiple zones/regions.
- Use managed services (GKE, Cloud SQL, BigQuery).
- Use load balancing (global or regional).
- Enable auto-healing and auto-scaling.
- Use Cloud Storage multi-region buckets.
- Implement retry and failover mechanisms.
- Define RTO (Recovery Time Objective) and RPO (Recovery Point Objective).
- Regular disaster recovery drills.

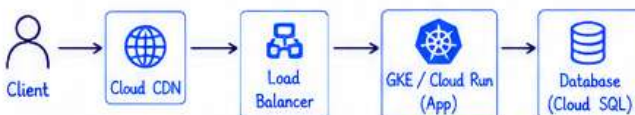
9. SECURITY BEST PRACTICES

- Follow least privilege access (IAM).
- Use service accounts instead of user keys.
- Enable VPC Service Controls.
- Use Cloud Armor for DDoS & WAF protection.
- Encrypt data at rest (CMEK / CSEK).
- Encrypt data in transit (HTTPS, TLS).
- Enable audit logs and monitoring.
- Use Secret Manager for sensitive data.
- Regularly rotate keys and credentials.
- Follow CIS benchmarks.

10. COST OPTIMIZATION CHECKLIST

- Use committed use discounts (1 or 3 years).
- Use preemptible/spot VMs for non-critical workloads.
- Right-size machine types.
- Delete idle resources (VMs, disks, IPs, snapshots).
- Use lifecycle policies for storage (Nearline / Coldline).
- Optimize BigQuery (partitioning, clustering).
- Monitor costs with Billing, Budgets and Alerts.
- Use labels to track costs by team/project.
- Regularly review and clean up unused resources.

11. USEFUL DIAGRAMS - QUICK REFERENCE



12. IMPORTANT TAKEAWAYS

- ☒ Design for scalability.
- ☒ Follow security best practices.
- ☒ Use managed services.
- ☒ Monitor, log and set alerts.
- ☒ Optimize costs.
- ☒ Be ready for real-world scenarios.
- ☒ Practice architecture diagrams.

Some
Concepts
Different Problems
Be Ready!

FINAL REVISION & CHEAT SHEET

Revise
Practice
Be Confident
You Can Do It !

1. GCP CORE SERVICES - QUICK RECALL

Category	Key Services
Compute	Compute Engine, Cloud Run, GKE, App Engine
Storage	Cloud Storage, Persistent Disk, Filestore, Local SSD
Databases	Cloud SQL, Firestore, Bigtable, Spanner, Memorystore
Networking	VPC, Cloud Load Balancing, Cloud CDN, Cloud DNS, Cloud Interconnect, Cloud VPN
Security	IAM, Cloud Armor, Secret Manager, KMS, IAP, Security Command Center
Data & Analytics	BigQuery, Dataflow, Pub/Sub, Dataproc, Looker, Data Studio
AI/ML	Vertex AI, AutoML, Vision API, NLP, Speech-to-Text
Management	Cloud Monitoring, Logging, Error Reporting, Trace, Profiler
Developer Tools	Cloud Build, Artifact Registry, Source Repositories, Cloud Deploy

2. IMPORTANT CONCEPTS

- Project → Top-level container for resources.
- Region → Geographical area (e.g., us-central1).
- Zone → Isolated data center within a region.
- Resource → Compute, storage, network, etc.
- IAM → Controls who can do what.
- VPC → Private network in Google Cloud.
- Service Account → Identity for applications.
- Managed Service → Fully managed by Google.
- Serverless → No server management (Cloud Run).
- Scalability → Auto-scale based on demand.
- High Availability → Multi-zone / multi-region.
- Fault Tolerance → Automatic recovery from failures.
- SLA → 99.9% or higher (depends on service).
- Pay-as-you-go → Pay only for what you use.
- Labels → Key-value pairs for organization.
- Quotas → Limits on resource usage.
- APIs → Access and manage GCP services.
- Cloud Console / gcloud / Terraform → Manage infra.

3. IAM ROLES - COMMON ONES

Role	Purpose
Owner	Full access to all resources
Editor	Modify resources (no IAM changes)
Viewer	Read-only access
Compute Admin	Manage Compute Engine resources
Storage Admin	Manage Cloud Storage resources
Kubernetes Admin	Manage GKE clusters
Security Admin	Manage security (IAM, KMS, etc.)
Service Account User	Use service accounts
Billing Admin	Manage billing and budgets
Custom Role	Fine-grained access (least privilege)

4. CHOOSE THE RIGHT SERVICE

Requirement	Recommended Service
Run a web app (no servers)	Cloud Run
Run containers (managed)	GKE
Simple backend (PaaS)	App Engine
Relational database	Cloud SQL
NoSQL document store	Firestore
Large scale analytics	BigQuery
Real-time data streaming	Pub/Sub + Dataflow
Static website hosting	Cloud Storage + CDN
In-memory caching	Memorystore (Redis)
Global load balancing	Cloud Load Balancing
DDoS & WAF protection	Cloud Armor
Machine learning platform	Vertex AI
Hybrid connectivity	Cloud VPN / Interconnect
Monitoring & logging	Cloud Monitoring + Logging

5. PRICING & COST TIPS

- Use committed use discounts (1 or 3 years).
- Use preemptible/spot VMs for non-critical workloads.
- Delete unused resources (VMs, disks, snapshots).
- Right-size machine types (e2, n2, etc.).
- Use lifecycle policies for storage (Nearline/Coldline).
- Optimize BigQuery queries (partitioning, clustering).
- Set up budget alerts and cost reports.
- Use labels to track project/team costs.
- Use cost calculator before deployment.
- Regularly review and clean up resources.
- Use autoscaling to avoid over-provisioning.
- Use free tier for learning and small projects.

6. SECURITY BEST PRACTICES

- Follow least privilege access (IAM).
- Use service accounts instead of user keys.
- Enable VPC Service Controls.
- Use Cloud Armor for DDoS & WAF protection.
- Encrypt data at rest (CMEK / CSEK).
- Encrypt data in transit (HTTPS, TLS).
- Enable audit logs and monitoring.
- Use Secret Manager for sensitive data.
- Regularly rotate keys and credentials.
- Use IAP for secure access to web apps.
- Follow CIS benchmarks.
- Use Binary Authorization for containers.
- Enable Security Command Center.
- Regularly review permissions and access logs.

7. HIGH AVAILABILITY & DR

- Deploy across multiple zones/regions.
- Use managed services (GKE, Cloud SQL, BigQuery).
- Use load balancing (global or regional).
- Enable auto-healing and auto-scaling.
- Use multi-region storage buckets.
- Implement automated backups.
- Define RTO (Recovery Time Objective).
- Define RPO (Recovery Point Objective).
- Test disaster recovery plan periodically.

8. COMMON INTERVIEW QUESTIONS

- Difference between VPC and VPN?
- When to use Cloud Run vs GKE?
- How does load balancing work in GCP?
- How to secure a web application on GCP?
- How to store and analyze large datasets?
- How to reduce costs in GCP?
- What is the default network in GCP?
- How does IAM work?
- How to design a multi-region architecture?
- How to handle sudden traffic spikes?
- How to monitor and troubleshoot an application?
- Real-world scenario based questions.

9. USEFUL gcloud COMMANDS

Task	Command
List projects	gcloud projects list
Set project	gcloud config set project <PROJECT_ID>
List compute instances	gcloud compute instances list
List storage buckets	gcloud storage buckets list
List GKE clusters	gcloud container clusters list
Check IAM policy	gcloud projects get-iam-policy <PROJECT_ID>
View logs	gcloud logging read "resource.type=..."
Deploy to Cloud Run	gcloud run deploy <SERVICE_NAME>
Build Docker image	gcloud builds submit --tag <IMAGE_NAME>
Check current config	gcloud config list

10. KEY DIFFERENCES (QUICK)

Feature	Compute Engine	Cloud Run	GKE	App Engine
Type	IaaS	Serverless	Managed K8s	PaaS
Manage servers	Yes	No	No	No
Scaling	Manual	Automatic	Automatic	Automatic
Use case	Full control	Containers	Containerized apps (K8s)	Web apps (PaaS)

11. REAL WORLD USE CASES

- E-commerce: GKE + Cloud SQL + CDN + Armor
- Media Streaming: Cloud Storage + CDN
- IoT: Pub/Sub + Dataflow + BigQuery
- ML Platform: Vertex AI + BigQuery
- Hybrid Cloud: Anthos
- Enterprise App: App Engine / Cloud Run
- Analytics Dashboard: Looker + BigQuery
- Disaster Recovery: Multi-region + automated backups

12. FINAL CHECKLIST

- ✓ Understand core services and use cases.
- ✓ Know IAM roles and security best practices.
- ✓ Be ready for scenario-based questions.
- ✓ Practice architecture diagrams.
- ✓ Focus on cost optimization.
- ✓ Revise gcloud commands.
- ✓ Do hands-on practice in Google Cloud Console.
- ✓ Review quotas, limits, and SLA.
- ✓ Stay updated with latest GCP features.
- ✓ Be confident and keep learning !

Small
Steps
Big Career
!